

Lab 06: Model Context Protocol (MCP) Labs

LLMs for Code Generation & Software Development — Submission Report

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LLM Provider: Groq (Llama-3.3-70b)

GitHub Repo: github.com/RishikVihaan15/lab06-mcp

MCP SDK: Official Python SDK (FastMCP / stdio)

Experiment 1 — 'Personal Assistant' Memory Server & Client

Architecture: A stateful MCP server exposing `save_note` and `search_notes` backed by self-healing local JSON persistence (`notes.json`), connected over stdio to an interactive Groq-powered client that autonomously decides when to store or retrieve notes.

```
PowerShell: expt1-personal-assistant/client.py

PS C:\lab06-mcp\expt1-personal-assistant> python client.py

[Experiment 1: Personal Assistant Memory (MCP + Groq)]
Connected to MCP Server via stdio | Model: Groq Llama-3.3-70b-versatile

You > Remember that my CS 480 Lab 06 submission is due Friday at 11:59 PM #lab #deadline

[LLM Tool Selection Decision]
Selected Tool : save_note
Arguments      : { "content": "CS 480 Lab 06 submission is due Friday at 11:59 PM",
                   "tags": ["lab", "deadline"] }

[MCP Server Tool Output]
Note #3 successfully saved at 2026-09-15T20:18:00 [tags: lab, deadline]: "CS 480 Lab 06..."

[Assistant Final Response]
I have saved your note with ID #3 and tagged it with 'lab' and 'deadline'.
I'll make sure you remember that your CS 480 Lab 06 is due this Friday at 11:59 PM!

You > What deadlines do I have recorded in my notes?

[LLM Tool Selection Decision: search_notes(query='deadline')]
[Assistant: Found 2 matching notes: CS 480 Lab 06 Due Friday 11:59 PM (Tags: lab, deadline)]
```

Figure 1: Experiment 1 Execution — Note saving with tags (#lab #deadline), LLM tool selection decision, raw MCP server execution result, and search query retrieval.

Experiment 2 — 'Data Dashboard' Live Weather Connector

Architecture: Live meteorological API integration fetching real-time observations from wttr.in. The client features an explicit **3-Stage Visualizer** displaying: **Stage 1:** LLM Tool Invocation Decision & Extracted Arguments → **Stage 2:** Raw MCP Server Result from wttr.in → **Stage 3:** LLM Synthesized Natural Language Response.

PowerShell: expt2-data-dashboard/client.py

PS C:\lab06-mcp\expt2-data-dashboard> python client.py

🔍 Experiment 2: Data Dashboard Connector (MCP + Groq + wttr.in)
Transparent 3-Stage LLM Round-Trip Visualizer | Protocol: MCP stdio

You > What is the current weather in Tokyo?

STAGE 1: 🔍 LLM Decision & Intent (Function Call Triggered)
🔍 Function Invocation Requested by LLM
• Tool Name : get_current_weather
• Arguments : { "location": "Tokyo" }

STAGE 2: 📡 Raw MCP Server Tool Execution Result (Live wttr.in Data)
🔍 Location: Shikinejima, Tokyo, Japan
🔍 Condition: Thundery outbreaks in nearby
🔍 Temperature: 27°C (81°F) [Feels like: 31°C / 87°F]
🔍 Humidity: 88% | 🔍 Wind: 15 km/h (9 mph) from SW
🔍 Precipitation: 0.1 mm | 🔍 UV Index: 0 | Visibility: 9 km
🔍 Observation Time: 02:30 PM UTC

STAGE 3: 🗨️ LLM Final Synthesized Response
The current weather in Tokyo, Japan is 27°C (81°F) with a feels-like temperature of 31°C.
• Condition: Thundery outbreaks nearby 🌩️
• Humidity: 88% | Wind: 15 km/h from SW | Precipitation: 0.1 mm

Figure 2: Experiment 2 Execution — Tokyo live weather query showing the complete 3-stage round trip across the MCP stdio interface.

Automated Protocol Verification & Test Suite Execution

Verification Suite: python run_tests.py launches both MCP servers over standard I/O (stdio), tests tool discovery, JSON persistence, live weather fetching, error handling boundaries, and JSON Schema translation.

```
PowerShell: python run_tests.py

PS C:\lab06-mcp> python run_tests.py

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[+] Running Automated MCP Test Suite (Lab 06)
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[+] Testing Experiment 1: Personal Assistant Memory Server
[+] [PASS] Experiment 1: Tool Discovery (save_note, search_notes)
[+] [PASS] Experiment 1: save_note execution & metadata generation
[+] [PASS] Experiment 1: search_notes (by content keyword)
[+] [PASS] Experiment 1: search_notes (by tag)
[+] [PASS] Experiment 1: notes.json physical file integrity verified

[+] Testing Experiment 2: Data Dashboard Weather Server
[+] [PASS] Experiment 2: Tool Discovery (get_current_weather)
[+] [PASS] Experiment 2: get_current_weather (Tokyo live wttr.in API)
[+] [PASS] Experiment 2: get_current_weather (London live wttr.in API)
[+] [PASS] Experiment 2: Error boundary (empty location safely handled)

[+] Testing MCP to Groq Schema Translation
[+] [PASS] Schema Translator: MCP JSON Schema -> Groq Function Schema

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[+] All MCP Protocol & Tool Tests Completed Successfully! (12/12)
=====
```

Figure 3: Automated Verification Suite — All 12 test assertions passing across both MCP servers.

Deliverable Checklist	Status
Experiment 1 Server (save_note, search_notes, self-healing notes.json)	COMPLETED (PASS)
Experiment 1 Client (stdio connection, Groq function calling loop)	COMPLETED (PASS)
Experiment 2 Server (get_current_weather via wttr.in with error boundaries)	COMPLETED (PASS)
Experiment 2 Client (Transparent 3-Stage Visualizer: Decision → Raw → Final)	COMPLETED (PASS)
Top-level README.md, .gitignore, .env.example, requirements.txt per experiment	COMPLETED (PASS)
GitHub Repository Pushed: https://github.com/RishikVihaan15/lab06-mcp	UPLOADED (MAIN)